LUCAS FENG

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EDUCATION

The Johns Hopkins University, Whiting School of Engineering — Expected May 2029
Bachelor of Science, Mechanical Engineering — Cumulative GPA: 3.88/4.0
Relevant Coursework: Statics and Mechanics of Materials, Mechanical Engineering Thermodynamics

SKILLS

- **Design and Simulation:** SolidWorks, Fusion 360, Onshape
- **Languages:** Python, C++, Java
- **Manufacturing:** CNC lathe/mill, welding, sand casting, sheet metal forming, FDM/resin 3D printing, laser cutting

EXPERIENCE

Johns Hopkins Neuroengineering and Biomedical Instrumentation Laboratory — May 2026 – Present
Undergraduate Researcher in Tactile Sensing

- Lead the mechanical development of two tactile sensing systems and established the lab's calibration methodology for multi-axis force sensors using dead weight loading and regularized least squares fitting.
- Designed the sensor housing for a Velostat-based instrumented insole that captures plantar loading and joint-level kinematics, mentoring the high school researchers who built it, for out-of-clinic recovery tracking after knee replacement.

Johns Hopkins Baja SAE — Sep 2025 – Present
Drivetrain Engineer

- Diagnosed, disassembled, and rebuilt the drivetrain of a competition off-road vehicle, collaborating with a 5-person sub team to restore full transmission functionality under competition constraints.
- Designed and fabricated a custom photogate-based acceleration measurement system, enabling data-driven drivetrain tuning and performance optimization across varied terrain conditions.

A-Level Capital — Feb 2026 – Present
Associate

- Invest in early-stage, capital-efficient companies with ties to Johns Hopkins as part of a student-run fund managing $5M in assets under management.
- Sourced founders and led introductory calls to assess team, product, and traction; conducted due diligence on market, competition, and financials, building Excel models to project ARR growth and market size, then synthesizing findings into investment memos.

EriCAD — Oct 2025 – Jul 2026
Co-Founder

- Developed an AI powered GD&T automation platform that predicts manufacturing tolerances from part geometry, functional requirements, manufacturing capability, and cost constraints.
- Led the platform from concept to a fully working demonstration over a four-month build cycle; selected and funded from the Johns Hopkins Forge Incubator.

PROJECTS

3-Axis Force Sensor, *Johns Hopkins Neuroengineering and Biomedical Instrumentation Laboratory*

- Designed and built a $120 3-axis force sensor using four load cells as a low-cost alternative to commercial systems costing more than $2,000.
- Developed the mechanical architecture, calibration procedure, and signal processing pipeline to accurately recover three-dimensional force measurements from four load cell outputs.
- Validated the sensor against a commercial reference across more than 340,000 samples, beating published low-cost sensors and submitted to IEEE IROS 2026 workshops.

Visuo-Tactile Bumps, *Johns Hopkins Neuroengineering and Biomedical Instrumentation Laboratory*

- Designed silicone tactile bump unit cells that convert shear forces into measurable pressure patterns for large area robotic tactile skin through CAD, nonlinear contact FEA, mold design, fabrication, and hardware integration.
- Performed a three-stage design optimization comparing bump height and profile geometry using measured pressure response to identify a design that achieved 98% of the maximum shear gain with only 68% of the deflection.
- Built the experimental validation pipeline with 50 grip and slip trials using synchronized tactile and vision data to support a multimodal transformer that predicts slip 500 ms before failure.